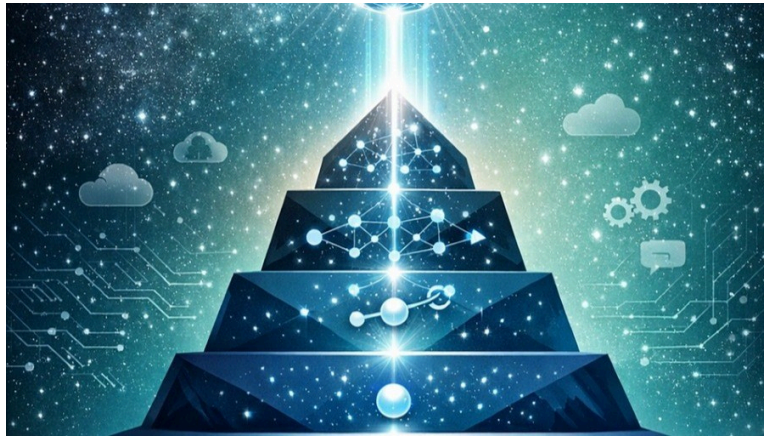


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Agent Capability Maturity Model (aCMM)

Not All AI Agents Are the Same — So Why Do We Treat Them Like They Are?



Suresh Chande ✓

Principal Product Manager, Data & Insights across Copilot, Agent platforms & Microsoft 3 IQs(Fabric, Foundry, Work IQ)...



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We're no longer dealing with one kind of agent — we're dealing with an entire ecosystem. We have task-automation agents that handle workflows, healthcare agents that triage symptoms or coordinate care, and embodied agents inside humanoids, drones, and mobile robots. Calling all of them simply "agents" is like calling every animal in your home a "pet." Technically true, but not remotely specific. Saying "I have a pet" tells you nothing. Saying "I have pet named Thor, a Golden Retriever" tells you almost everything that matters.

That's why understanding what we mean by "agent" is no longer optional. The differences between these systems shape expectations, risks, governance, and how we deploy them in the real world. My argument is simple: if we want to build and scale agents responsibly, we need a clearer language — and that's where the Agent Capability Maturity Model (aCMM) comes in.

We're all using the word "AI agent" — but we often mean completely different things.

A **task-automation agent** that files tickets, an **LLM tool-orchestrator**, a **multi-agent coordination system**, and an **autonomous mobile robot** can all be labeled "agents" ... yet their capabilities, risks, and governance needs are nowhere near comparable.

To solve this, several attempts are made to define “agent maturity” ladders, including:

- **Salesforce’s Agentic Maturity Model** (copilots → orchestration → multi-agent) [5]
- **Agent Maturity Model** by Samet Kilictas, Technical Architect @ IBM (Helper → Integrator → Planner → Boss → Team) [3]
- **5 Levels of Agentic AI Intelligence** by [Shubha Pant](#) and [Mahesh Viswanathan @ Cisco](#) (autonomy ladder) [4]
- **Agentic AI Maturity** by [James M. Sims](#), Founder and Consultant @ [Cognition Consulting Group](#) (ad hoc → self-evolving systems) [6]
- **AI Agent Taxonomy** by [Daniel W. Rasmus](#) @ Serious Insight (argue agents can’t be reduced to a single ladder) [1]

These frameworks are genuinely helpful — they give us shared language and a roadmap.

But here’s the issue I keep seeing in real deployments: a **single linear “level” breaks down almost immediately.**

Teams building agents often see capabilities accelerate unevenly. An agent may advance quickly in areas like **tool use**, **memory**, or **planning**, while other dimensions — especially **autonomy** or **learning** — are deliberately constrained for safety, compliance, or operational reasons. This asymmetry becomes even more pronounced with **embodied agents** such as robots and drones, or agents deployed in regulated environments. And as LLM-based agents gain self-improving behaviors, their capabilities no longer fit neatly into any single linear maturity ladder.

That’s why I’m proposing a more robust approach: the **Agent Capability Maturity Model (aCMM)** — a multi-dimensional capability-profile framework that evaluates agents across **eight distinct dimensions** instead of collapsing everything into a single maturity score.

Autonomy · Reasoning · Learning · Memory · Tool Use · Collaboration · Embodiment · Domain Alignment



Agent Classification

Each dimension is scored from 0–5, producing a **capability fingerprint** — a visual shape that makes trade-offs explicit and gives engineering, product, security, and governance teams a shared, evidence-based way to discuss what an agent can (and cannot) do. Over time, this model can also expand to include **domain-specific dimensions** for sectors like healthcare, manufacturing, or finance, where specialized capabilities and constraints meaningfully shape how agents must be evaluated.

When you compare **software agents in digital workspace**, **embodied agents operating in physical space**, **medical and clinical AI agents**, and **mobile robotic or drone agents**, you see completely different capability fingerprints — and suddenly the word “agent” stops being a single bucket and becomes a full spectrum of systems with very different risks, expectations, and governance needs.

The article walks through how to score agents, identify bottlenecks, and pair higher autonomy with stronger controls so organizations can scale agent programs responsibly.

Interested to know more you can read the full article: [\[aCMM\]: Agent Capability Maturity Model - A Unified Framework for Evaluating Modern AI Agents | by Suresh Chande | Mar, 2026 | Medium](#)

If you prefer to Listen, go to the Copilot podcast here: [Agent Capability Framework](#) or see this video created below:

aCMM - An Agent Capability Maturity Model

I'd love to hear your feedback and curious if this resonates with you or you can think of Dimensions and scaling worth considering to this framework.

#AI #Agents #AgenticAI #aCMM #ArtificialIntelligence #Autonomy
#AIEngineering #AIGovernance #FutureOfWork #LLM

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6. [The Five Levels of Agentic AI Maturity – Cognition Consulting Group](#) By James M. Sims, Founder and Consultant

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One dimension that may eventually need consideration is not "consciousness" itself — since we still lack a scientific definition for that — but the degree of persistent self-modeling and identity continuity an agent maintains over time.

...more

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1mo ...

[Sotirios Makrygiannis](#) love that.. Glad to learn you see ways to grow this, we certainly need a good framework standard to be developed in the open including industry, academia, government/NGOs to agree on agent taxonomy and classification so represent them in a way there is no confusion. Let's collaborate to elaborate it further. ...more

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Andrea Mansfield · 3rd+

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1mo ...

Suresh, yes. The word "agent" is starting to carry too many different systems at once.

A task assistant, a tool-orchestrating agent, and an embodied system ask very different things of the person using them. Each one changes wha ...more

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1mo ...

[Andrea Mansfield](#) thanks for your reminder and note you are absolutely on spot and this is so critical to put a pin on what an Agent is and is not . I can't agree more on that point as there is a need to align on what an Agent is. I believe we are at the third generation of Agent Evolution and entering into the fourth soon , every phase the de ...more

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